Department of Chemical Engineering, Chulalongkorn Univ

MTJA CO_2-to-methanol pilot · operator training simulator (teaching re-issue)

DOCUMENT NO. : R601-IN-CE-001

DOCUMENT TITLE : CAUSE AND EFFECT DIAGRAM

UNIT : R-601 pilot trickle-bed reactor, CO_2 + H_2 to methanol with ethanol

SOURCE : Re-drawn for teaching from 2407-IN-CE-001 rev A2

REVISION : A

This sheet reproduces the structure and the settings of the issued cause-and-effect diagram so that it can be used with the simulator. Company names, logos and people of the source document are not reproduced. Trip delays are the values used by the simulator (assumed: not given on the source document). Where the simulator departs from the source (TAL-602 re-based, band heaters tripped with E-601), the note column says so.

Rev.	Date	Revision status	Prepared	Checked	Approved
A	24-Oct-26	Issued for teaching	A. Soottitantawat		

CAUSES

EFFECTS

ITEM	TAG	SERVICE	DIRECTION	SETTING	UNIT	DELAY s	NOTE	XV-101	FV-101	XV-201	FV-201	XV-301	P-301	XV-401	FV-401	E-601 HEATER	XV-601	XV-602	XV-603	Flashing lamp (amber)	Flashing lamp (red)	Audible alarm
TRIP ACTION								C	C	C	C	C	TP	O	FO	TP	C/O	C/O	C/O	A	A	A
001	ESD-SW	Emergency shutdown switch	Tripped			0	1	X	X	X	X	X	X	X	X	X	C	C	O		X	X
002	PAL-203	H_2 supply pressure low	Low	27	bar g															X		X
003	PALL-203	H_2 supply pressure low-low	Low-Low	24	bar g	2		X	X	X	X	X	X	X	X	X	C	C			X	X
004	PAL-103	CO_2 supply pressure low	Low	27	bar g															X		X
005	PALL-103	CO_2 supply pressure low-low	Low-Low	24	bar g	2		X	X	X	X	X	X	X	X	X	C	C	O		X	X
006	PAL-403	N_2 supply pressure low	Low	24	bar g															X		X
007	PAHH-301	Ethanol supply pressure high-high	High-High	62	bar g	1		X	X	X	X	X	X	X	X	X	C	C	O		X	X
008	PAH-301	Ethanol supply pressure high	High	59	bar g															X		X
009	PALL-301	Ethanol supply pressure low-low	Low-Low	50	bar g	2		X	X	X	X	X	X	X	X	X	C	C			X	X
010	PAL-301	Ethanol supply pressure low	Low	53	bar g															X		X
011	FAL-101	CO_2 flow low	Low	95 % SP	L/min															X		X
012	FAH-101	CO_2 flow high	High	105 % SP	L/min															X		X
013	FAL-201	H_2 flow low	Low	95 % SP	L/min															X		X
014	FAH-201	H_2 flow high	High	105 % SP	L/min															X		X
015	FAL-301	Ethanol flow low	Low	95 % SP	L/min															X		X
016	FAH-301	Ethanol flow high	High	105 % SP	L/min															X		X
017	FAHH-301	Ethanol flow high-high	High-High	110 % SP	L/min															X		X
018	FAL-401	N_2 flow low	Low	< 7.5	L/min		2													X		X
019	LAH-301	Ethanol storage level high	High	80	%															X		X
020	LAL-301	Ethanol storage level low	Low	20	%															X		X
021	PAL-601	R-601 inlet pressure low	Low	29	bar g															X		X
022	PALL-601	R-601 inlet pressure low-low	Low-Low	27	bar g	2		X	X	X	X	X	X	X	X	X	C	C			X	X
023	PAH-601	R-601 inlet pressure high	High	32	bar g															X		X
024	PAHH-601	R-601 inlet pressure high-high	High-High	33	bar g	1		X	X	X	X	X	X	X	X	X	C	C	O		X	X
025	PAH-608	D-601 pressure high	High	32	bar g															X		X
026	PAHH-608	D-601 pressure high-high	High-High	33	bar g	1		X	X	X	X	X	X	X	X	X	C	C	O		X	X
027	TAH-601	E-601 temperature high	High	158	°C															X		X
028	TAHH-601	E-601 temperature high-high	High-High	167	°C	2		X	X	X	X	X	X	X	X	X	C	C	O		X	X

NOTES

1. Trip actions: C closed, TP tripped, O opened, FO fully opened, C/O closed or opened as set per row (columns XV-601/602/603), A alarm.
2. A trip row (shaded) closes XV-101/201/301 and FV-101/201, stops P-301, opens XV-401 and FV-401 (N_2 purge), trips E-601.
3. Simulator: the band heaters E1/E2/E3 trip with E-601 (assumed; not on the source).
4. Low-low trips marked "RUN only" in the simulator are bypassed while the SIS is in READY (start-up); see the SIS tab.
5. Note 1 (source): operating range and setting to be defined (FAL-401, FALL-701).
6. Note 3 (source): independent level switch LS-605 on D-601 (LAHH-605).
7. TAL-602 143 °C / TALL-602 135 °C assume an all-vapour inlet; the simulator alarms TAL-602 at 110 °C (assumed, trickle-bed inlet ≈ 127 °C).
8. Flow alarms at ±5 % of the controller set point act only in AUTO/CAS.
9. Delay: simulator trip delay in seconds (assumed; the source gives none).
10. Findings for review (HAZOP R601-SA-HZP-001): PAL-203/103 at 27 bar g below the 30 bar g reactor; TT-604 has no alarm; XV-601 is both the LIC-601 valve and the LALL-601 final element; one PLC for control and trips.

CAUSE AND EFFECT DIAGRAM · R-601 PILOT TRICKLE-BED REACTOR

Client Department of Chemical Engineering, Chulalongkorn University
Project MTJA CO_2-to-methanol pilot · operator training simulator (teaching re-issue)
Document no. R601-IN-CE-001 rev A · sheet 1 of 2
Source After 2407-IN-CE-001 rev A2 (13-Nov-25), rows 001–054; P&ID 2407-PE-PID-001 A6; R601-PID-001 rev A

REV	DATE	STATUS	ORIGINATOR	CHECKER	APPROVER
A	24-Oct-26	Issued for teaching	A. Soottitantawat		

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EFFECTS

ITEM	TAG	SERVICE	DIRECTION	SETTING	UNIT	DELAY s	NOTE	XV-101	FV-101	XV-201	FV-201	XV-301	P-301	XV-401	FV-401	E-601 HEATER	XV-601	XV-602	XV-603	Flashing lamp (amber)	Flashing lamp (red)	Audible alarm
TRIP ACTION								C	C	C	C	C	TP	O	FO	TP	C/O	C/O	C/O	A	A	A
029	TAL-602	R-601 inlet temperature low	Low	143	°C															X		X
030	TALL-602	R-601 inlet temperature low-low	Low-Low	135	°C																X	X
031	TAH-602	R-601 inlet temperature high	High	158	°C															X		X
032	TAHH-602	R-601 inlet temperature high-high	High-High	167	°C	2		X	X	X	X	X	X	X	X	X	C	C	O		X	X
033	TAH-605	R-601 outlet temperature high	High	158	°C															X		X
034	TAHH-605	R-601 outlet temperature high-high	High-High	167	°C	2		X	X	X	X	X	X	X	X	X	C	C	O		X	X
035	LALL-601	D-601 level low-low	Low-Low	10	%	5		X	X	X	X	X	X	X	X	X	C	C	O		X	X
036	LAL-601	D-601 level low	Low	20	%															X		X
037	LAH-601	D-601 level high	High	60	%															X		X
038	LAHH-601	D-601 level high-high	High-High	80	%	5		X	X	X	X	X	X	X	X	X	O	O	O		X	X
039	LALL-602	D-602 level low-low	Low-Low	10	%	5		X	X	X	X	X	X	X	X	X	C	C			X	X
040	LAL-602	D-602 level low	Low	20	%															X		X
041	LAH-602	D-602 level high	High	60	%															X		X
042	LAHH-602	D-602 level high-high	High-High	80	%	5		X	X	X	X	X	X	X	X	X	C	O			X	X
043	LAH-603	T-601 level high	High	80	%															X		X
044	LAL-603	T-601 level low	Low	20	%															X		X
045	FAL-701	Chilled water supply flow low	Low	63.65	L/min															X		X
046	FALL-701	Chilled water supply flow low-low	Low-Low	60.3	L/min	5	2	X	X	X	X	X	X	X	X	X	C	C	O		X	X
047	TAL-609	E-602 outlet temperature low	Low	8	°C															X		X
048	TAH-609	E-602 outlet temperature high	High	32	°C															X		X
049	TAHH-609	E-602 outlet temperature high-high	High-High	34	°C	5		X	X	X	X	X	X	X	X	X	C	C	O		X	X
050	PALL-902	Instrument air header pressure low-low	Low-Low	0.9	bar g	2	1	X	X	X	X	X	X	X	X	X					X	X
051	PAL-902	Instrument air header pressure low	Low	1.2	bar g															X		X
052	PAH-902	Instrument air header pressure high	High	4.8	bar g															X		X
053	PAHH-902	Instrument air header pressure high-high	High-High	5.1	bar g	2	1	X	X	X	X	X	X	X	X	X					X	X
054	LAHH-605	D-601 level high-high (independent LS-605)	High-High	85	%	5	3	X	X	X	X	X	X	X	X	X	O	O	O		X	X
055		(spare)																				
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NOTES

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Document no. R601-IN-CE-001 rev A · sheet 2 of 2
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